Battle For Tomorrow

Technology is enabling the movement of warfare from a blood-drenched killing affair to something more precisely-targeted and yet more powerful

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War is a sad reality: has been teh same in the past and will continue to remain so in future. It is anybody's conjecture what war will be like, say, five decades from now. It will not involve the loss of lives in direct combat: we would have passed well beyond that stage. But what can it be about? Will it be about the destruction of physical targets via what the Internet will be at that time? Or will it be entirely cyber-warfare... attacks solely on information systems?

Whatever the case, the fact is that technology has decisively entered the battlefield, just as it has entered the sports arena. It makes it less fun and more serious business, but war is fun only when you're watching a movie!

Here, then, are some things that have happened in recent years, which point to the inexorable rush of technology into combat of all sorts. We focus on the communications and networking aspect, because that's where it's happening most. We have good reason to believe that what's happening today is indicative of what will be tomorrow.

The Networked Soldier

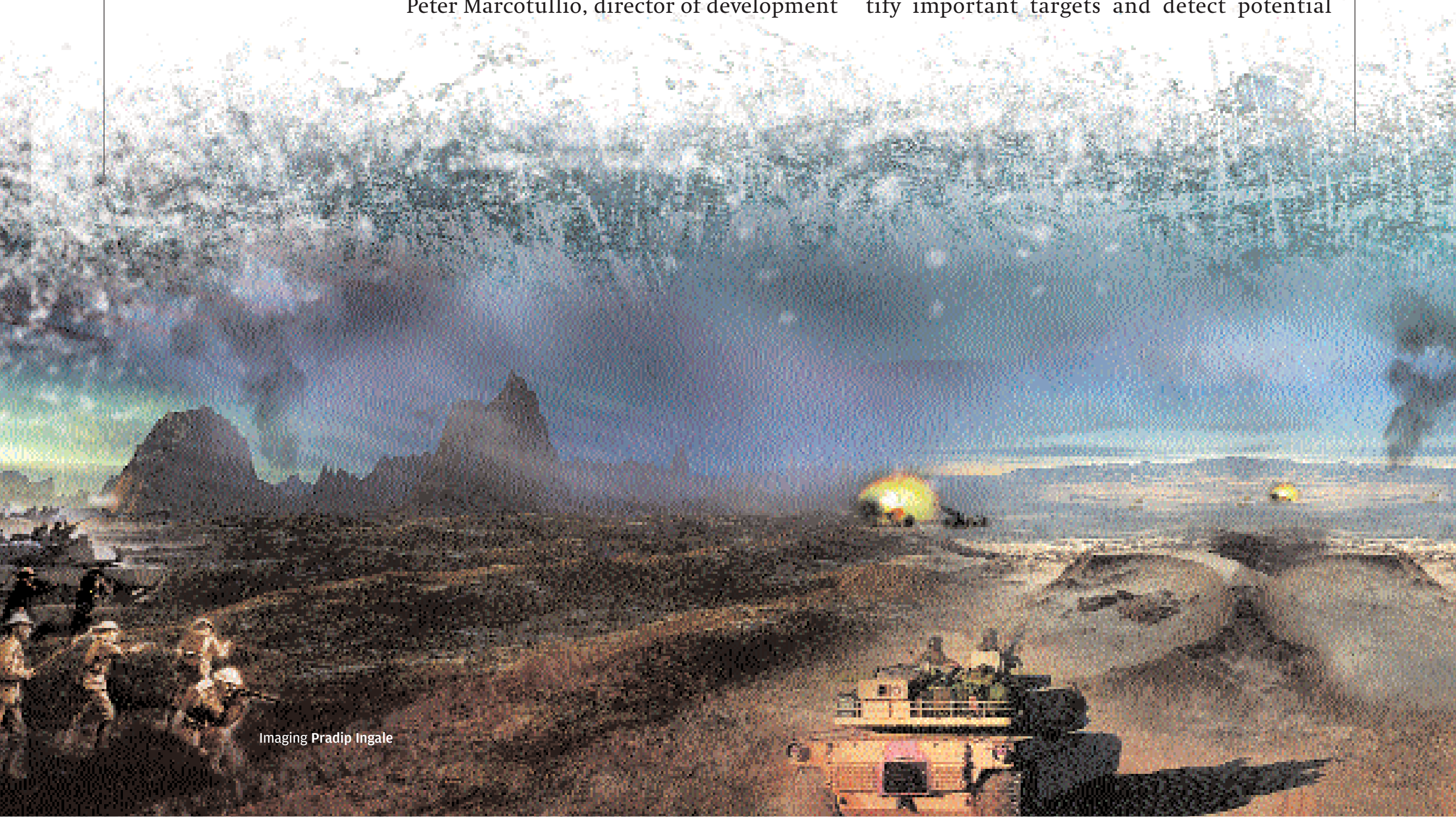
In the US Army's vision called the Future Combat System, networks will acquire the capability of being set up on the fly to adapt to the progress of a battle.

Peter Marcotullio, director of development at SRI International, the research centre working with the DoD (the US Department of Defense) on ad hoc network development, says, "Each person is a network with routing capability to everyone else. Think of cascading networks, all IP-based, that are dynamic and self-configured as the troops advance."

Sensors, commanders and soldiers will all be linked to a dynamic network that will adapt to the situation and satisfy the information demands of the battlefield. This vision is not far from being realised. By the end of this decade, major armed forces across the globe will have sensors and communication gear that will make such connectivity possible.

In addition to conventional weapons, the soldier of the future will have the necessary equipment to make him a network-enabled "smart warrior." Using his helmet-mounted drop-down eyepiece, he will see a virtual, GPS-linked display showing his current location and that of friendly and hostile units in the area. He will be able to switch the display to watch live video feeds from UAVs (Unmanned Aerial Vehicles) coming via a secure data link. His rifle-mounted TWS (Thermal Weapon Sight) will not only help him identify enemy positions in the dark, but also pass on the same information to allied units on the network.

A soldier is just a part of a platoon or company of similarly-equipped soldiers. The soldier utilises data fed in by sensors like UAVs, aircraft, satellites, and other sensors to identify important targets and detect potential



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